

Course Introduction

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

Andrew Begel

Prof. Andrew Begel

- Tenured Associate Professor at Carnegie Mellon University
- Affiliate Professor at University of Washington, Seattle
- Formerly Principal Researcher at Microsoft Research
- PhD in Computer Science from University of California, Berkeley
- BS & MEng from Massachusetts Institute of Technology
- <https://andrewbegel.com>

My Organizations



VariAbility

ACCESSIBILITY IN COMPUTATION LABORATORY



SOUTHERN GREAT LAKES REGION
Neurodiversity at Work Hub

My Mission

To create inclusive workplaces where *all* people, especially those with disabilities and the neurodivergent, can be successful, *without discrimination*.

Teaching Assistant



Neeta
Khanuja

Classroom Culture Settings

- Honor our different ways of being.
 - Diversity is fundamental to building and maintaining an equitable and inclusive campus community.
- Leverage our classroom space and shared items within it as you like.
- Treat one another like professionals. Give them the benefit of the doubt.
- Let people approach problems in their own way
 - If you get frustrated, don't be afraid to ask questions
- Almost every problem can be solved with communication.
 - Without communication, problems grow and grow.
- Each of us is responsible for creating a safer, more inclusive environment.

Join the class Slack

- Get out your computer.
- Join our Slack: <https://bit.ly/accessibility-slack-invite>



Introductions

In the Slack channel, **#in-class**, write a post with 5 parts.

1. Preferred Name
2. Preferred Pronouns
3. Major
4. Year at CMU
5. What does accessibility mean to you?
(write one sentence or phrase)

Next, we'll go around the room and have everyone introduce themselves.

Intro to Accessibility and Assistive Technology

- Why are we here?
- Philosophy on this course

Course Topics

- Disabilities: Models, Policy, Advocacy
- Web, Mobile, Desktop, Wearable Accessibility
 - Tech Stack
 - Design
 - Guidelines
- ATs for Sight, Hearing, Speech, Mobility, Cognition, Communication
- AI for Accessibility, Responsible AI
- Domains: Software Development, Enjoying Life, Civic Infrastructure, Employment, Education, Fabrication

Definitions

- Disability
- Models of Disability
- Functional Impairments
- Accessibility
- Assistive Technology (AT)

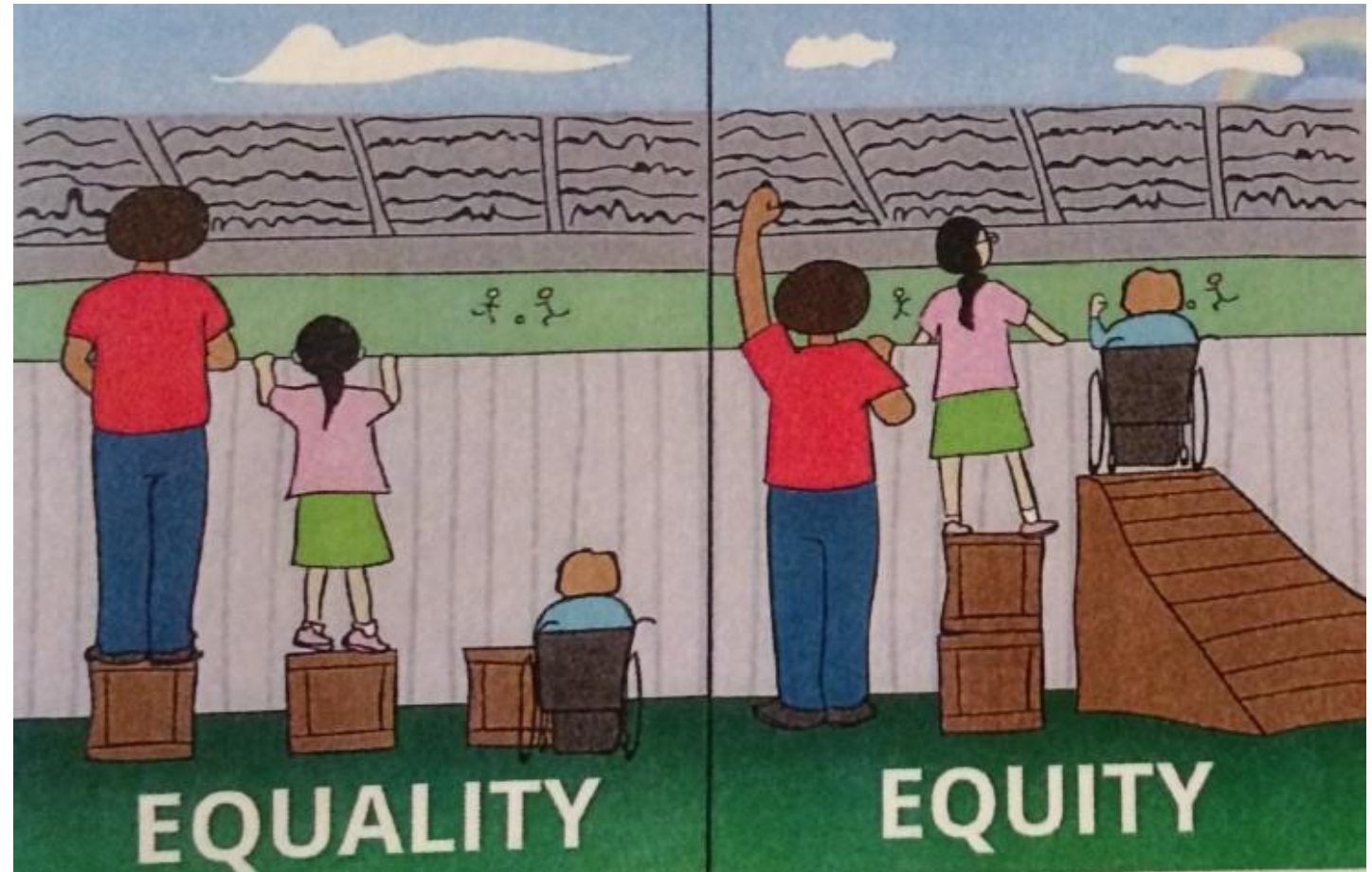
Perspectives on Disability and Access

- Medical
- Social
- Legal
- Cultural
- Identity-based
- Intersectional
- Advocate's

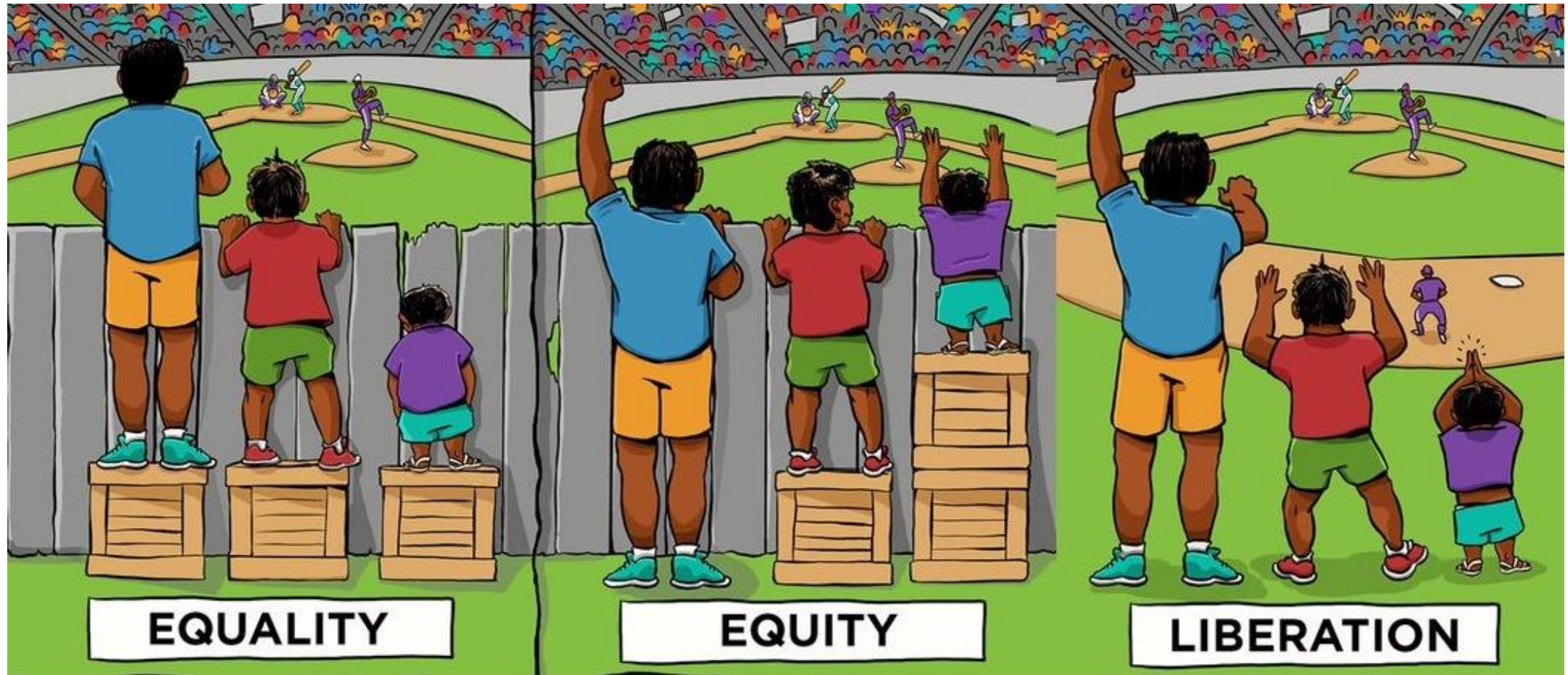
Aim for Inclusion

When you do not intentionally, deliberately, and proactively include, you will unintentionally exclude.

– Joe Gerstandt



Shoot for Liberation



What *non-technological*
assistive technologies (ATs)
have you heard of?

Post as many ATs as you can think of to
#in-class.

What modern *technological*
assistive technologies (ATs)
have you heard of?

Post as many ATs as you can think of to
#in-class.

Screen Readers for Visual Impairments



JAWS



VoiceOver



NVDA



Talkback

Assistive Tech for Visual Impairments



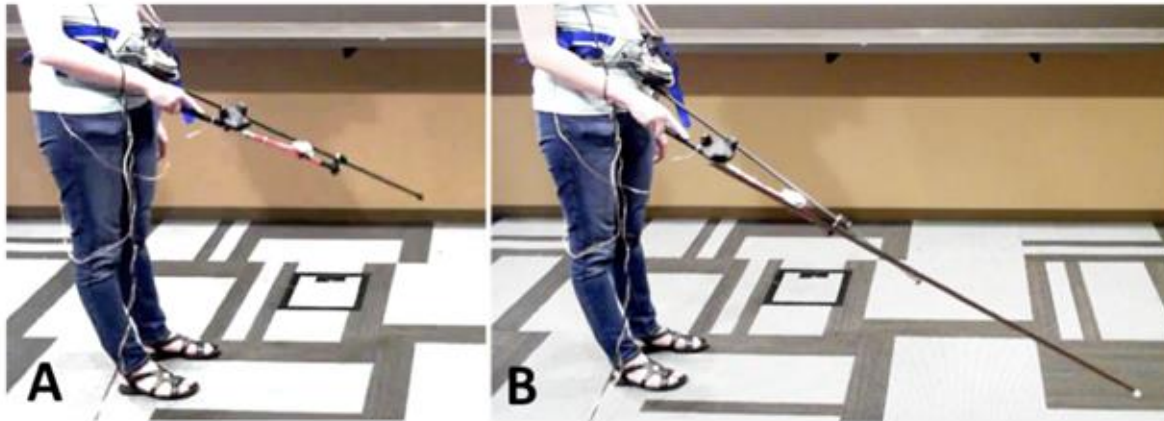
SeeingAI



SoundScape



Code Jumper



Canetroller



VizWiz

Assistive Tech for Hearing Impairments



SoundWatch



Apple iOS Sound Recognition

Syllabus

Available online at <https://cmu-intro-to-accessibility.github.io/syllabus/>

Prerequisites

- Technically, there are no prerequisites.
- Be willing...
 - To experiment with using ATs.
 - To use AI to design and implement ideas for ATs.
 - To communicate with real people to learn about their experiences with disability and ATs.

Waitlist

We will add as many students as we can.
Waitlists are processed in FIFO order.

If you're on the waitlist for 05332/05632, sign up for 17332/17732. It's the same class!

If you are stuck on the waitlist, we recommend attending the first couple of weeks of class and attempting the first homework.

Assessment

- Individual Homework - 50%
 - Group Project - 30%
 - Participation - 20%
-
- See the syllabus for more details on points assigned to each homework and project.

Teamwork

- Projects will be done done in teams of 4-5 students.
- Teams will be self-organized and stay together for the entirety of the class. Choose your teammates wisely.
- Every team should mix designers, engineers, and researchers. You will need a mix of each for a successful project.

Teamwork

- Being able to address team issues in software projects is one of the core learning objectives of this class.
- Guidance on teamwork, reflection, and conflict resolution will be provided throughout the semester.
- We expect your effort to address any team issues before asking instructors to step in.

Class Participation

- To participate in class, use a body part to visually flag our attention or write in Slack channel #in-class.
- We expect in-person attendance at every class.
 - There is no hybrid option for this course.

Missed Participation

- You may miss up to 3 participation activities without losing points.
 - For any reason. Really, I don't need to know.
 - Official CMU activities do not impact these 3 if you tell us at least 24 hours in advance.
 - Emergencies (e.g., car accident, apocalypse, deadly virus) do not impact these 3.
 - Talk to us in private about any extended absences.
- You may only contribute to a participation activity if you are physically present in class. Doing so while absent is an academic integrity violation.
- You may not assign someone else's name to a participation activity who is not physically present in class. This is an academic integrity violation.

Reading Participation

- Every class will have a set of readings assigned.
 - You should pick one to read before that class begins.
- Be prepared to answer questions in class about the main idea and supporting evidence from the reading.
- Be careful when using AI to summarize a reading for you.
 - Our learning goal is to have the information in the reading in *your* head, not the AI's.
 - AIs will easily hide important details from you. Their sycophancy will incentivize them to hide their lies until you confront them.

Grading

- Regrade policy
 - Regrade requests can be submitted via Gradescope.
 - The **regrade period is open for one week** after grades have been released for a particular assignment.
 - After one week, you may no longer request a regrade or correction to the grading of an assignment.
- All digital submissions must be made *accessible* (as described in HW1)

Late Individual Work

- Turn in individual homework on time.
 - 1 day late: 10% penalty
 - 2 days late: 20% penalty
 - 3 days late: 50% penalty
 - 4+ days late: 100% penalty
- Late work will still receive feedback.

Late Team Work

- Due to heavy emphasis on planning & teamwork, there will be no credit for late team project assignments.
- Communicate with your team early if you won't get your work done on time.
- Every team should have a plan to mitigate the risk of a team member failing to complete their portion of the assignment.

Time Management

- 12 hours a week
 - 3 hours in class
 - 9 hours outside of class

Writing + AI

- You may use generative AI technologies, such as ChatGPT, Claude, or CoPilot, unless the assignment question specifically prohibits it.

AI Caveats

1. You must clearly indicate the use of such technology (and name the tool) in every homework question or source code file where an AI tool was used as a form of attribution, just as you would if you had taken help from a friend;
2. Anything you submit that contains ideas or verbatim quotes from external sources must be cited using APA style. You **must** also check that the reference really exists.
3. You are liable for factually inaccurate answers or unspecific rambling produced by AI tools; it is your responsibility to edit AI-produced content before submitting it for class purposes.

Accommodations

- If you need any accommodations, please work with the Disability Resources Office.
 - They will send me an email with your requests.
 - I'll email you back to discuss how we work with them in this class.
- If you suspect that you may have a disability and would benefit from accommodations, contact access@andrew.cmu.edu.

Help us learn more about you

<https://bit.ly/accessibility-teamwork-survey>

